

Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration
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Practical Answer Submission

Practical Sheet No: 07

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Question 1

- i) $2^{10}=1024$ (1024-1)
New processes created = 1023 (excluding original parent)
In fork () new child processors are being created
- ii) Child process executes function_A()
fork() returns:
- 0 → to child
 - >0 → to parent
- iii) Child's PID is printed by the parent
- In parent: pid > 0 → contains child's PID
- In child: pid == 0
- iv) Ensures parent waits for all child processes

Question 2

- i) Total no of processors= 2^N
 $= 2^5$
32 processors
- ii)

```
int main() {  
    int i;  
    pid_t pid;  
  
    for (i = 0; i < 3; i++) {  
        pid = fork();  
  
        if (pid == 0) {  
            CPU(); // child executes  
            return 0;  
        }  
    }  
}
```

```

    }
}

// parent waits
for (i = 0; i < 3; i++) {
    wait(NULL);
}

return 0;
}

```

Question 3

PARENT: value= 40

Question 4

i) Total processes:

$$2^4 = 16$$

in for loop it iterates 4 times and fork() is executed four times. In fork it creates child processors.

ii) Value= 45

Value= 15

- Both parent & child execute printf
- Order may vary, but usually:
- Child prints first (parent waits)

Question 5

- i) Stores thread IDs
- ii) Creates 25 threads
- iii) Function executed by each thread
- iv) Makes main thread wait for all threads